

Agastya Sharma

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TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, C, C++

Frameworks and Libraries: React, React Native, Next.js, Redux, Tailwind CSS, Node.js, Flask, FastAPI, PyTorch

DevOps and Cloud: Docker, AWS, CI/CD, GitHub Actions, Redis, Vercel

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Arts in Computer Science, Co-op (GPA: 3.90 / 4.00)

2025 – 2030

- President's Scholarship of Distinction; Michael Egan Entrance Scholarship in Computer Science

WORK EXPERIENCE

Sylphia Consulting Inc.

Sep 2025 – Present

Full-Stack Software Developer

Toronto, ON

- Led an end-to-end full-stack project using **Next.js**, and **PostgreSQL** on the backend delivering a production-ready web application within 4 months with a user base of **50+ users**
- Created a nutrition **mobile application** using with **Expo**, implementing camera capture, real-time data updates via **Supabase**, a dashboard, and user profile management for healthcare providers
- Developed backend AI pipelines using **Gemini Vision** for image recognition, **PyTorch-based depth-estimation models**, and **custom AI agents/assistants** for automation and data processing
- Implemented **RESTful APIs** using **Node.js**, **FastAPI**, and **Flask** on 3 different products
- Redesigned product and software using foundational **UI/UX** principles using **Figma** to improve user experience

FEATURED PROJECTS

Nutrition Application — Next.js, TypeScript, Python, C, PostgreSQL

Jan 2026

- Developed full-stack application using **Python** and **C** for backend processing and system optimization
- Analyzed and **designed system architecture** for electrical component integration and data flow
- Built and tested **RESTful APIs** with focus on reliability and performance in production environment

Music Insights Dashboard — Flask, Python, JavaScript, SQLite

Sep 2025

- Built a full-stack music analytics dashboard using **Flask**, **Jinja**, and **JavaScript**, integrating Spotify's **OAuth 2.0** flow to securely authenticate users and fetch personalized listening data
- Implemented a content-based recommendation engine by **extracting** and **vectorizing** song attributes, computing similarity scores to generate personalized track recommendations after Spotify deprecated audio feature **endpoints**
- Developed interactive **UI** components with **Bootstrap CSS** and JavaScript to visualize top artists, tracks, playlists, and listening trends across multiple time windows

Oil Price Prediction Model — Python, C++, NumPy, PyTorch

Jan 2025

- Developed end-to-end machine learning pipeline using Python and C++ for computational efficiency
- Implemented neural network architectures demonstrating understanding of complex systems and optimization
- Analyzed model performance with strong attention to detail using RMSE and MAE metrics

3D ASCII Renderer — Python, C, NumPy, Linear Algebra

Dec 2024

- Built real-time rendering engine in Python and C applying mathematical transformations
- Developed and optimized code for embedded system-like constraints with limited computational resources
- Demonstrated curiosity and aptitude to learn by implementing complex graphics algorithms independently

AWARDS

Incubator Hacks Winner; International Diamond Challenge State Qualifiers; Two-Time Provincial and Regional DECA Champion in ENT and ETDM; PGCC Continental Finalist; Tech Under 20 Winner; Top 25% CSMC, Euclid, and CCC